MILITARY SPECIFICATION

POWDER, MOLDING, PBX 9010

1. SCOPE

1.1 This specification covers one type of PBX 9010, Molding Powder, a high explosive mixture composed of RDX and a desensitizer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, For Testing Purposes.

MILITARY

MIL-R-00398 - RDX.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

F7548644 - Box, Packing for High Explosive Assembly Details Packing and Marking.
F7548645 - Carton, Packing, Reuseable, Collapsible for High Explosives Assembly Details, Packing and Marking.
F7548646 - Carton, Packing, Reuseable, Collapsible, Alternative for High Explosives, Assembly Details, Packing and Marking.

Source: https://assist.dla.mil -- Downloaded: 202640-22T03:01Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (latest revision)".

3. REQUIREMENTS

3.1 Material.-The molding powder shall be a mixture of RDX surrounded by a homogenous desensitizer coating. The constitutent materials used in the manufacture of the molding powder shall comply with the following requirements.

Constituent material:	Conforming to:
RDX	MIL-R-00398, Type B, Class A or Type B, Class G and Type B, Class E.
Desensitizer	Kel-F-Elastomer 3700, Product of Minnesota Mining and Manufacturing Company, St. Paul 6, Minnesota; or approved substantial equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:01Z
Check the source to verify that this is the current version before use.

3.2 Granulation.-The molding powder shall be supplied in the form of free-flowing particles which shall comply with the following granulation requirements when tested as specified in 4.4.1.

U.S. Sieve No.	Cumulative Percent Retained
4	5 Maximum (max.)
50	95 Minimum (min.)
80	100

3.3 Volatiles.-The volatile content shall be 0.10 percent max., when tested as specified in 4.4.2.

3.4 Composition.-The analytical composition of the molding powder shall be as follows when tested as specified in 4.4.3.

Constituent	Percent by weight
RDX	90.0 ± 0.3
Desensitizer	10.0 ± 0.3

(a) The specified percentage of RDX shall consist of 75 percent by weight of Type B; Class A or Type B, Class G, RDX and 25 percent by weight of Type B, Class E, RDX.

3.5 Color.-The color of the molding powder shall be white to off-white at the time of acceptance, when tested as specified in 4.4.4.

3.6 Bulk density.-The bulk density of the molding powder shall be 0.80 grams (gm.) per cubic centimeter (c.c.) min. when tested as specified in 4.4.5.

3.7 Foreign matter.-The molding powder shall be free of metal inclusions, wooden particles, and other foreign matter when tested as specified in 4.4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the

Source: https://assist.dla.mil -- Downloaded: 202640-22T03:01Z
Check the source to verify that this is the current version before use.

right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.3 and 4.4 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.3 and 4.4 and other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.3 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Quantity of product inspected.

c. Results obtained for all inspection performed.

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:01Z
Check the source to verify that this is the current version before use.

d. Specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure (see 4.1.2), this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and, depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

Source: https://assist.dla.mil -- Downloaded: 20260-22T03:01Z
Check the source to verify that this is the current version before use.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of one or more batches of molding powder, produced by one manufacturer, in accordance with the same specification, or the same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of molding powder that has been subjected to the same unit chemical or physical mixing process, intended to make the final product homogeneous. Each lot shall consist of:

a. RDX from not more than one lot from one manufacturer.

b. Desensitizer from not more than one lot from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2.1 Box and cartons (prior to closing filled container) (see dwgs. F7548644, F7548645 and F7548646).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Bag liner missing or torn	Visual
102.	Top of bag liner not double folded	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:01Z
Check the source to verify that this is the current version before use.

Minor: AQL 0.40 percent
201. Bag liner pulled too tight not allowing slack ..... Visual

4.3.2.2 Box and cartons (after closing container) (see dwgs. F7548644, F7548645 and F7548646).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Sealing strips or strapping, as applicable, missing	Visual
102.	Marking missing or illegible	Visual
Minor:	AQL 1.00 percent	
201.	Sealing strips or strapping, as applicable, not properly applied	Visual
202.	Marking not properly applied	Visual

4.2.3 Testing.
4.2.3.1 Determination of physical and chemical properties (see 3.2 through 3.6) - Major defects.
4.2.3.1.1 Sampling by lot.-A random sample of 9 containers shall be selected from each lot. When lots somprise 9 containers or less, each container shall be selected.
4.2.3.1.2 Preparation of composite sample.-A 1.0 ± 0.2 ounce primary sample of the molding powder shall be removed from each of the 9 containers in order to equal 9 ounces. If there are less than 9 containers, equal primary samples in sufficient quantity to equal 9 ounces shall be removed from each container. The individual primary samples shall then be combined in order to form a homogeneous composite sample of 9 ounces. The determination shall be performed as specified in 4.3. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:01Z
Check the source to verify that this is the current version before use.

4.3 Test methods and procedures.

4.3.1 Granulation.-A 100 ± 0.1 gm. specimen of the dry sample shall be placed on the specified nest of sieves, complying with Specification RR-S-366, properly superimposed and assembled with a bottom pan. The nest of sieves shall be covered and shaken for 3 minutes by means of a mechanically operated sieve shaker. The portions retained or passed by the various screens shall be weighed and the results calculated to a cumulative percentage basis.

4.3.2 Volatiles.-The volatile content of the powder shall be determined as follows: Weigh into a 100 by 15mm Petri dish a specimen of approximately 10 gm. of the sample. Record the weight to the nearest milligram. Shake dish gently to distribute the sample evenly over the bottom of the dish. Place dish in vacuum oven and dry at 60 degrees Centigrade (°C.) until the sample reached constant weight (approximately 2 hours). Remove dish and cool in a desiccator to room temperature. Weigh and calculate volatiles as follows:

$$\text{Percent volatiles} = \frac{A \times 100}{B}$$

where:

A = loss in weight of the specimen

B = original weight of the specimen

4.3.3 Composition.-Analytical composition of the finished powder shall be determined by the following procedures.

4.3.3.1 Determination of desensitizer.

4.3.3.1.1 Introduction.-This procedure is based on the solvent extraction of RDX with concentrated nitric acid. The desensitizer is determined by difference.

4.3.3.1.2 Special apparatus.-The following special apparatus shall be used in this determination:

a. Soxhlet extraction apparatus (250 ml. flask).
b. Mantle, heating, hemispherical, 250 ml.
c. Crucible, fritted glass, coarse porosity, 20 mm, tall form.
d. Transformer, variable.

4.3.3.1.3 Procedure.-Obtain the sample to be analyzed by conventional quartering techniques. If the agglomerate size is greater than 16 mesh, gently crush the sample so that all particles will pass a USS-16 sieve. Weigh to the nearest milligram approximately 1 gram (gm.) of powder into a tared 20 millimeter (mm.) tall form, fritted glass crucible. Place the crucible in a suitable Soxhlet extraction apparatus. Add approximately 125 milliliters (ml.) of concentrated nitric acid (specific gravity 1.42) into the flask of the Soxhlet apparatus and insert an anti-bump rod. Assemble the complete Soxhlet apparatus on a ring stand. Turn on the circulating water to the condenser. Adjust the voltage to the heating

Source: https://assist.dla.mil -- Downloaded: 20261O-22T03:01Z
Check the source to verify that this is the current version before use.

mantle so that after the acid starts to condense and drop into the crucible, a rate of approximately 45 drops per minute is obtained. Allow extraction to continue for 4 hours or until sample appears transparent (further extraction is not harmful). After the extraction is complete allow the system to cool. Remove the crucible containing the residue and allow the remaining liquid to drain from the crucible. Wash the crucible and contents with distilled water at least three times. Place the crucible and residue in a vacuum oven for 2 hours or until it reaches constant weight. Weigh the residue and calculate percent desensitizer as follows:

$$\text{Percent desensitizer} = \frac{A}{B} \times 100$$

where:

A = weight of residue
B = weight of specimen

4.3.3.2 Determination of RDX.-Calculate the RDX content as follows: percent RDX = 100 percent minus percent desensitizer content (see 4.4.3.1.2).

4.3.4 Color.-Color of the molding powder shall be determined by visual inspection. Slight variation of color between particles or marked discoloration of isolated particles shall not be considered a cause for rejection of the lot.

4.3.5 Determination of bulk density.

4.3.5.1 Apparatus.-A plywood box having inside dimensions 4 by 4 by 4 inches, padded on top and bottom with 1/4 inch foam rubber, and having a hole in a hinged top large enough to fit a 250 c.c. graduated cylinder, shall be employed in this determination.

4.3.5.2 Procedure.-The bulk density of the powder shall be determined as follows: Accurately weigh a 200 gm. specimen of the sample. Pour the powder into a 250 c.c. graduated cylinder. Enclose the cylinder in the box. Lift the protruding end of the cylinder until the bottom flange of the cylinder contacts the top of the box. Drop the cylinder. Again lift and drop the cylinder until 20 drops have been made at the rate of one per second. Read the volume of the compressed powder in the cylinder. Calculate bulk density in gms, per c.c., as follows:

$$\text{Bulk density} = \frac{A}{B}$$

where:

A = weight of powder (gms.)
B = volume of compressed powder (c.c.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:01Z
Check the source to verify that this is the current version before use.

4.3.6 Foreign matter.-The molding powder shall be inspected visually to assure that it is free of foreign matter.

5. PREPARATION FOR DELIVERY

5.1 Packing.-PBX 9010 Molding Powder shall be packed and marked in accordance with dwgs. F7548644, F7548645 or F7548646 as applicable. Molding powder from not more than one lot shall be packed in one container. Boxes or cartons shall be packed full to a uniform net weight not to exceed 50 or 60 pounds as shown on the applicable drawing.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Packing requirements.
c. When required by the consignee, the contractor control data of the determination for each individual batch in the blended lot, for the following, using test procedures as specified in 4.4.

(1) Granulation (3) Composition (5) Bulk Density
(2) Volatiles (4) Color (6) Foreign Matter

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensinggthe holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:01Z
Check the source to verify that this is the current version before use.